

Navigation Plugins

There are a number of plugin interfaces for users to create their own custom applications or algorithms with. Namely, the costmap layer, planner, controller, behavior tree, and behavior plugins. A list of all known plugins are listed here below for ROS 2 Navigation. If you know of a plugin, or you have created a new plugin, please consider submitting a pull request with that information.

This file can be found and edited under `sphinx_docs/plugins/index.rst` . For tutorials on creating your own plugins, please see [Writing a New Costmap2D Plugin](#), [Writing a New Behavior Tree Plugin](#), [Writing a New Controller Plugin](#), [Writing a New Planner Plugin](#), [Writing a New Behavior Plugin](#), or [Writing a New Navigator Plugin](#).

Behavior-Tree Navigators

Plugin Name	Creator	Description
NavigateToPoseNavigator	Steve Macenski	Point-to-point navigation via a behavior tree action server
NavigateThroughPosesNavigator	Steve Macenski	Point-through-points navigation via a behavior tree action server
CoverageNavigator	Steve Macenski	Complete coverage navigation (Cartesian or GPS) via a BTs

Costmap Layers

Plugin Name	Creator	Description
Voxel Layer	Eitan Marder-Eppstein	Maintains persistent 3D voxel layer using depth and laser sensor readings and raycasting to clear free space
Range Layer	David Lu	Uses a probabilistic model to put data from sensors that publish range msgs on the costmap
Static Layer	Eitan Marder-Eppstein	Gets static <code>map</code> and loads occupancy information into costmap
Inflation Layer	Eitan Marder-Eppstein	Inflates lethal obstacles in costmap with exponential decay
Obstacle Layer	Eitan Marder-Eppstein	Maintains persistent 2D costmap from 2D laser scans with raycasting to clear free space
Spatio-Temporal Voxel Layer	Steve Macenski	Maintains temporal 3D sparse volumetric voxel grid with decay through sensor models
Non-Persistent Voxel Layer	Steve Macenski	Maintains 3D occupancy grid consisting only of the most sets of measurements
Denoise Layer	Andrey Ryzhikov	Filters noise-induced standalone obstacles or small obstacles groups
Plugin Container Layer	Alexander Yuen	Combines the different costmap layers specified under this layer in order populate the same costmap with different isolated combinations of costmap layers

Costmap Filters

Plugin Name	Creator	Description
Keepout Filter	Alexey Merzlyakov	Maintains keep-out/safety zones and preferred lanes for moving
Speed Filter	Alexey Merzlyakov	Limits maximum velocity of robot in speed restriction areas
Binary Filter	Alexey Merzlyakov	Enables binary (boolean) mask behavior to trigger actions.

Controllers

Plugin Name	Creator	Description	Drivetrain support
DWB Controller	David Lu!	A highly configurable DWA Implementation with plugin interfaces	Differential, Omnidirectional, Legged
TEB Controller	Christoph Rösmann	A MPC-like controller suitable for ackermann, differential, and holonomic robots.	Ackermann, Legged, Omnidirectional, Differential
Regulated Pure Pursuit	Steve Macenski	A service / industrial robot variation on the pure pursuit algorithm with adaptive features.	Ackermann, Legged, Differential
MPPI Controller	Steve Macenski Aleksei Budyakov	A predictive MPC controller with modular & custom cost functions that can accomplish many tasks.	Differential, Omni, Ackermann
Rotation Shim Controller	Steve Macenski	A "shim" controller to rotate to path heading before passing to main controller for tracking.	Differential, Omni, model rotate in place
Graceful Controller	Alberto Tudela	A controller based on a pose-following control law to generate smooth trajectories.	Differential, Omni, Legged
Vector Pursuit Controller	Black Coffee Robotics	A controller based on the vector pursuit algorithm useful for high speed accurate path tracking.	Differential, Ackermann, Legged,

Planners

Plugin Name	Creator	Description	Drivetrain support
NavFn Planner	Eitan Marder-Eppstein & Kurt Konolige	A navigation function using A* or Dijkstras expansion, assumes 2D holonomic particle	Differential, Omnidirectional, Legged
SmacPlannerHybrid (formerly <i>SmacPlanner</i>)	Steve Macenski	A SE2 Hybrid-A* Implementation using either Dubin or Reeds-shepp motion models with smoother and multi-resolution query, Cars, car-like, and ackermann vehicles. Kinematically feasible.	Ackermann, Differential, Omnidirectional, Legged
SmacPlanner2D	Steve Macenski	A 2D A* implementation Using either 4 or 8 connected neighborhoods with smoother and multi-resolution query	Differential, Omnidirectional, Legged
SmacPlannerLattice	Steve Macenski	An implementation of State Lattice Planner using pre-generated minimum control sets for kinematically feasible planning with any type of vehicle imaginable. Includes generator script for Ackermann, diff, omni, and legged robots.	Differential, Omnidirectional, Ackermann, Legged, Arbitrary / Custom
ThetaStarPlanner	Anshumaan Singh	An implementation of Theta* using either 4 or 8 connected neighborhoods, assumes the robot as a 2D holonomic particle	Differential, Omnidirectional

Smoother

Plugin Name	Creator	Description
Simple Smoother	Steve Macenski	A simple path smoother for infeasible (e.g. 2D) planners
Constrained Smoother	Matej Vargovcik & Steve Macenski	A path smoother using a constraints problem solver to optimize various criteria such as smoothness or distance from obstacles, maintaining minimum turning radius
Savitzky-Golay Smoother	Steve Macenski	A path smoother using a Savitzky-Golay filter to smoothen the path via digital signal processing to remove noise from the path.

Behaviors

Plugin Name	Creator	Description
Clear Costmap	Eitan Marder-Eppstein	A service to clear the given costmap in case of incorrect perception or robot is stuck
Spin	Steve Macenski	Rotate behavior of configurable angles to clear out free space and nudge robot out of potential local failures
Back Up	Brian Wilcox	Back up behavior of configurable distance to back out of a situation where the robot is stuck
Wait	Steve Macenski	Wait behavior with configurable time to wait in case of time based obstacle like human traffic or getting more sensor data
Drive On Heading	Joshua Wallace	Drive on heading behavior with configurable distance to drive
Assisted Teleop	Joshua Wallace	Assisted Teleop behavior that scales teleop commands to prevent collisions.

Waypoint Task Executors

Plugin Name	Creator	Description
WaitAtWaypoint	Fetullah Atas	A plugin to execute a wait behavior on waypoint arrivals.
PhotoAtWaypoint	Fetullah Atas	A plugin to take and save photos to specified directory on waypoint arrivals.
InputAtWaypoint	Steve Macenski	A plugin to wait for user input before moving onto the next waypoint.

Goal Checkers

Plugin Name	Creator	Description
SimpleGoalChecker	David Lu!	A plugin check whether robot is within translational distance and rotational distance of goal.
StoppedGoalChecker	David Lu!	A plugin check whether robot is within translational distance, rotational distance of goal, and velocity threshold.
PositionGoalChecker	Prabhav Saxena	A plugin check whether robot is within translational distance of goal, without requiring rotational convergence.

Progress Checkers

Plugin Name	Creator	Description
SimpleProgressChecker	David Lu!	A plugin to check whether the robot was able to move a minimum distance in a given time to make progress towards a goal
PoseProgressChecker	Guillaume Doisy	A plugin to check whether the robot was able to move a minimum distance or angle in a given time to make progress towards a goal

Behavior Tree Nodes

Action Plugin Name	Creator	Description
Back Up Action	Michael Jeronimo	Calls backup behavior action
Drive On Heading Action	Joshua Wallace	Calls drive on heading behavior action
Assisted Teleop Action	Joshua Wallace	Calls assisted teleop behavior action
Clear Entire Costmap Service	Carl Delsey	Calls clear entire costmap service
Clear Costmap Except Region Service	Guillaume Doisy	Calls clear costmap except region service
Clear Costmap Around Robot Service	Guillaume Doisy	Calls clear costmap around robot service
Compute Path to Pose Action	Michael Jeronimo	Calls Nav2 planner server
Smooth Path Action	Matej Vargovcik	Calls Nav2 smoother server
Follow Path Action	Michael Jeronimo	Calls Nav2 controller server
Navigate to Pose Action	Michael Jeronimo	BT Node for other BehaviorTree.CPP BTs to call Navigation2 as a subtree action
Reinitialize Global Localization Service	Carl Delsey	Reinitialize AMCL to a new pose
Spin Action	Carl Delsey	Calls spin behavior action
Wait Action	Steve Macenski	Calls wait behavior action
Truncate Path	Francisco Martin	Modifies a path making it shorter
Truncate Path Local	Matej Vargovcik	Extracts a path section around robot
Planner Selector	Pablo Iñigo Blasco	Selects the global planner based on a topic input, otherwise uses a default planner id
Controller Selector	Pablo Iñigo Blasco	Selects the controller based on a topic input, otherwise uses a default controller id
Goal Checker Selector	Pablo Iñigo Blasco	Selects the goal checker based on a topic input, otherwise uses a default goal checker id
Smoother Selector	Owen Hooper	Selects the smoother based on a topic input, otherwise uses a default smoother id
Progress Checker Selector	Steve Macenski	Selects the progress checker based on a topic input, otherwise uses a default progress checker id
Navigate Through Poses	Steve Macenski	BT Node for other BehaviorTree.CPP BTs to call Nav2's NavThroughPoses action
Remove Passed Goals	Steve Macenski	Removes goal poses passed or within a tolerance for culling old viapoins from path re-planning
Remove In Collision Goals	Tony Najjar	Removes goal poses that have a footprint or point cost above a threshold.
Compute Path Through Poses	Steve Macenski	Computes a path through a set of poses rather than a single end goal pose using the planner plugin specified
Compute Route	Steve Macenski	Computes a Route through a navigation graph and returns both a dense path and set of sparse route nodes and edges.
Compute And Track Route	Steve Macenski	Computes a Route as above, but also actively tracks progress and triggers route contextual semantic operations.
Cancel Control Action	Pradheep Padmanabhan	Cancels Nav2 controller server
Cancel BackUp Action	Pradheep Padmanabhan	Cancels backup behavior action
Cancel Spin Action	Pradheep Padmanabhan	Cancels spin behavior action
Cancel Wait Action	Pradheep Padmanabhan	Cancels wait behavior action
Cancel Route Action	Steve Macenski	Cancels ComputeAndTrackRoute action
Cancel Drive on Heading Action	Joshua Wallace	Cancels drive on heading behavior action
Cancel Assisted Teleop Action	Joshua Wallace	Cancels assisted teleop behavior action
Cancel Complete Coverage Action	Steve Macenski	Cancels compute complete coverage
Compute Complete Coverage Path Action	Steve Macenski	Calls coverage planner server
Get Pose From Path Action	Marc Morcos	Extracts a pose from a path
Dock Robot Action	Steve Macenski	Calls dock robot action
Undock Robot Action	Steve Macenski	Calls undock robot action
Concatenate Paths Action	Steve Macenski	Concatenates 2 paths together
Get Current Pose Action	Steve Macenski	Gets current pose to the blackboard
Append Goal Pose To Goals Action	Steve Macenski	Appends a goal pose to a goals vector
Extract Route Nodes To Goals Action	Steve Macenski	Converts Route Nodes to Goals
Get Next Few Goals Action	Steve Macenski	Obtains the next N goals in a goal vector
Toggle Collision Monitor Service	David Grbac	Calls toggle collision monitor service

Condition Plugin Name	Creator	Description
Goal Reached Condition	Carl Delsey	Checks if goal is reached within tol.
Goal Updated Condition	Altior Miguel Blanco	Checks if goal is preempted.
Global Updated Goal Condition	Joshua Wallace	Checks if goal is preempted in the global BT context
Initial Pose received Condition	Carl Delsey	Checks if initial pose has been set
Is Stuck Condition	Michael Jeronimo	Checks if robot is making progress or stuck
Is Stopped Condition	Tony Najjar	Checks if robot is stopped for a duration
Transform Available Condition	Steve Macenski	Checks if a TF transformation is available. When succeeds returns success for subsequent calls.
Distance Traveled Condition	Sarthak Mittal	Checks is robot has traveled a given distance.
Time Expired Condition	Sarthak Mittal	Checks if a given time period has passed.
Is Battery Low Condition	Sarthak Mittal	Checks if battery percentage is below a specified value.
Is Path Valid Condition	Joshua Wallace	Checks if a path is valid by making sure there are no LETHAL obstacles along the path.
Path Expiring Timer	Joshua Wallace	Checks if the timer has expired. The timer is reset if the path gets updated.
Are Error Codes Present	Joshua Wallace	Checks if the specified error codes are present.
Would A Controller Recovery Help	Joshua Wallace	Checks if a controller recovery could help clear the controller server error code.
Would A Planner Recovery Help	Joshua Wallace	Checks if a Smoother recovery could help clear the planner server error code.
Would A Smoother Recovery Help	Joshua Wallace	Checks if a Smoother recovery could help clear the smoother server error code.
Would A Route Recovery Help	Steve Macenski	Checks if a Route recovery could help clear the route server error code.
Is Battery Charging Condition	Alberto Tudela	Checks if the battery is charging.
Are Poses Near Condition	Steve Macenski	Checks if 2 poses are nearby to each other.
Is Pose Occupied Condition	Maurice Alexander Purnawan	Checks if a pose is occupied.

Decorator Plugin Name	Creator	Description
Rate Controller	Michael Jeronimo	Throttles child node to a given rate
Distance Controller	Sarthak Mittal	Ticks child node based on the distance traveled by the robot
Speed Controller	Sarthak Mittal	Throttles child node to a rate based on current robot speed.
Goal Updater	Francisco Martin	Updates the goal received via topic subscription.
Single Trigger	Steve Macenski	Triggers nodes/subtrees below only a single time per BT run.
PathLongerOnApproach	Pradheep Padmanabhan	Triggers child nodes if the new global path is significantly larger than the old global path on approach to the goal
GoalUpdatedController	Sophia Koffler	Ticks child node if the goal has been updated

Control Plugin Name	Creator	Description
Pipeline Sequence	Carl Delsey	A variant of a sequence node that will re-tick previous children even if another child is running
Recovery	Carl Delsey	Node must contain 2 children and returns success if first succeeds, If first fails, the second will be ticked. If successful, it will retry the first and then return its value
Round Robin	Mohammad Haghighipanah	Will tick 1 child until a result and move on to 1+1
Nonblocking Sequence	Alexander Yuen	A variant of a sequence node that will tick through the whole sequence even if a child returns running. On reticks of this control node, successful children will be ticked once again to prevent a stale state from being latched.
Persistent Sequence	Enjoy Robotics	A variant of a sequence node that exposes <code>current_child_idx</code> as a bidirectional port.
Pause Resume Controller	Enjoy Robotics	Controlled through service calls to pause and resume the execution of the tree.

Route Plugins

Edge Scorers

Plugin Name	Creator	Description
DistanceScorer	Steve Macenski	Scores an edge's length, optionally scaled by relative speed limits.
TimeScorer	Steve Macenski	Scores and edge traversal time using absolute speed limits or previous traversal times.
PenaltyScorer	Steve Macenski	Scores using a static semantic penalty.
SemanticScorer	Steve Macenski	Scores using stored semantic data regarding the edge and/or nodes.
StartPoseOrientationScorer	Alex Yuen	Scores based on the initial pose and start edge orientations.
GoalPoseOrientationScorer	Alex Yuen	Scores based on the goal pose and goal edge orientations.
DynamicEdgesScorer	Steve Macenski	Scores based on a dynamically set service cost and/or closure.

Route Operations

Plugin Name	Creator	Description
AdjustSpeedLimit	Steve Macenski	Adjusts robot speed limits using an edge's semantic data.
CollisionMonitor	Steve Macenski	Checks for collision in the immediate future while tracking a route.
TimeMarker	Steve Macenski	Records the traversal time for an edge in the edge's metadata.
ReroutingService	Steve Macenski	Triggers a reroute from an external server.
TriggerEvent	Steve Macenski	Triggers an event based on a configurable server name.

Graph File Parsers

Currently, only `geosjon` parsing is supported.